Free-Body Diagrams

U03
PRACTICE SET

NAME DATE PERIOD SCORE / 14

BEFORE YOU START — YOU ALREADY KNOW THIS

- Newton's first law: an object at rest, or moving at a constant speed in a straight line, has balanced forces.
- If an object is speeding up or slowing down, the forces are not balanced. The larger forces point the way it is speeding up, or opposite the way it moves if it is slowing down.

FORCE LABEL BANK

gravity (weight)
Fg

normal force
FN

applied force
Fapp

friction
f (fs static, fk kinetic)

tension
T (two ropes: TL, TR)

air resistance
drag

push of one object on another
F book on table

friction acts
along the surface, opposite the sliding (or the tendency to slide)

DIRECTIONS

Draw the object as a box. Draw one arrow for each force, starting at the box, pointing the way the force acts. Label each arrow from the bank. Make longer arrows for larger forces when you can tell. Do not draw an arrow for motion. Write subscripts small, as in F with a small g.

1. • WARM-UP

A box of books sits at rest on a level floor. Draw the free-body diagram for the box.

DRAW THE FREE-BODY DIAGRAM

1

2. • WARM-UP

A crate on smooth, frictionless ice is pushed horizontally to the right and speeds up. Ignore air resistance. Draw the free-body diagram for the crate.

DRAW THE FREE-BODY DIAGRAM

1

3. • WARM-UP

A dresser is pushed horizontally to the right across a rough wooden floor at a constant speed. Ignore air resistance. Draw the free-body diagram for the dresser.

DRAW THE FREE-BODY DIAGRAM

1

4. • PRACTICE

A hockey puck slides to the right across rough ice and slows to a stop. Nothing is pushing it. Ignore air resistance. Draw the free-body diagram for the puck while it is still sliding.

DRAW THE FREE-BODY DIAGRAM

1

5. • PRACTICE

A framed picture hangs freely, not touching the wall, from two ropes. One rope pulls up and to the left and the other pulls up and to the right, each at 50° above the horizontal. Draw the free-body diagram for the picture.

DRAW THE FREE-BODY DIAGRAM

1

6. • PRACTICE

A person stands on a bathroom scale in an elevator. The elevator is moving up and speeding up. Draw the free-body diagram for the person.

DRAW THE FREE-BODY DIAGRAM

1

7. • PRACTICE

A skydiver falls toward the ground and is still speeding up. Air resistance acts on her. Draw the free-body diagram for the skydiver.

DRAW THE FREE-BODY DIAGRAM

1

8. • CHALLENGE

A sled is pulled to the right across rough snow at a constant speed. The rope pulls at 30° above the horizontal. Ignore air resistance. Draw the free-body diagram for the sled.

DRAW THE FREE-BODY DIAGRAM

1

9. • CHALLENGE

A toolbox sits at rest on a rough ramp tilted 25° above the horizontal. Draw the free-body diagram for the toolbox.

DRAW THE FREE-BODY DIAGRAM

1

10. • CHALLENGE

A wooden block slides down a rough ramp tilted 25° above the horizontal and speeds up as it goes. Ignore air resistance. Draw the free-body diagram for the block.

DRAW THE FREE-BODY DIAGRAM

1

11. • CHALLENGE

A book rests on a table, and the table stands on a level floor. Nothing is moving. Draw two separate free-body diagrams: one for the book and one for the table.

DRAW TWO SEPARATE FREE-BODY DIAGRAMS

1	2
BOOK	TABLE

12. • MULTI-TOPIC

A block on a frictionless table is tied to a string. The string runs over a pulley at the right edge of the table to a hanging mass, and the mass falls and speeds up. Ignore the mass of the string and the pulley, and ignore air resistance. Draw two separate free-body diagrams: one for the block and one for the hanging mass.

DRAW TWO SEPARATE FREE-BODY DIAGRAMS

1	2
BLOCK	HANGING MASS